Hyperspectral Image Few-Shot Classification Network Based on the Earth Mover's Distance

Jiaxing Sun, Xiaobo Shen, and Quansen Sun

***Abstract*— Deep learning has achieved promising performance in hyperspectral image (HSI) classification. Training deep models usually requires labeling massive HSIs, which, however, is prohibitively time-consuming and expensive. To fill in the gap, this article proposes a novel meta-learning method for HSI few-shot classification that conducts HSI classification with a few labeled samples. Specifically, we introduce the Earth mover's distance (EMD) as a metric. The designed EMD metric learning module aims to calculate the similarity of paired embedding features by decomposing embedding features into a set of local representations. The EMD metric aims to find the optimal matching flows between local representations that have the minimum matching cost. Furthermore, we attempt to learn class prototype representation for each hyperspectral class using the EMD metric. The proposed network effectively learns general knowledge from base HSIs and transfers such knowledge to the classification of novel HSIs. We conduct HSI few-shot classification by training on three base HSIs and classification on three novel HSIs. Extensive experimental results on three novel HSI datasets demonstrate that the proposed model outperforms the existing state-of-the-art HSI methods, including two HSI few-shot methods.**

***Index Terms*— Deep learning, few-shot, hyperspectral image (HSI) classification, meta-learning, the Earth mover's distance (EMD).**

I. Introduction

HYPERSPECTRAL imaging technology cooperates with spectral technology to detect geometric space and spectral information of the target. Hyperspectral image (HSI) contains rich spatial and spectral information of the target, and thus HSI has been widely applied in some emerging areas such as agriculture [1], ecological science, geological science [2], surveillance, and military applications [3]. HSI classification, which aims to specify a unique category for each pixel, plays a crucial role in various HSI applications, e.g., detection [4] and tracking [5]. The past decade has witnessed the success of conventional machine learning methods, e.g., support vector machines (SVMs) [6], Markov random fields [7], and random forest [8]. Recently, Wu *et al.* [9] present a survey of state-of-the-art methods for processing remotely sensed big data such as HSI classification. Later deep learning is gradually showing its potential and superiority in hyperspectral feature representation. The convolutional neural network (CNN) [10], deep belief networks (DBNs) [11], and recurrent neural network (RNN) [12] have achieved appealing HSI classification performance. Deep methods usually require sufficient labeled samples in each class for training; however, collecting plenty of labeled HSIs is expensive and time-consuming. For example, the Indian Pines (IP) dataset only has 46 and 20 labeled samples in the alfalfa and oats classes, respectively. How to conduct HSI classification with limited labeled samples remains a challenging problem.

Semisupervised learning, which uses model assumptions on data distribution to explore unlabeled data, has been widely applied for the scarcity of hyperspectral samples. Some generative adversarial network (GAN)-based models, e.g., a deep convolutional GAN (DCGAN) [13] and a semisupervised framework for HSI data based on GAN (HSGAN) [14], have achieved qualified classification results. These methods either focus on exploring information from limited labeled samples with the aid of prior knowledge or use unlabeled data for information compensation. They only exploit information from the target HSI, which limits generalization of model and causes overfitting. The trained model fails to discriminate between new classes of the novel HSI. However, retraining a new classifier for each novel HSI is infeasible and wasteful in real applications.

Enlighteningly, humans can recognize new objects based on knowledge they have learned. Inspired by such idea, few-shot classification was proposed, where a classifier can recognize novel classes with the aid of only a few labeled samples per class. As a mainstream strategy in few-shot classification, meta-learning mechanism encourages model to learn abundant knowledge from base dataset and transfer knowledge to novel classification. Recently, metric learning approaches [15]–[17] have attracted constant concern in HSI classification. They usually design a feature embedding module to map all data into an embedding space and determine the distances between the paired embeddings with a metric function. For instance, based on [16], Liu *et al.* [18] compute Euclidean distances between labeled samples and samples to be classified. Gao *et al.* [19] extend [17] to learn relationship scores between paired hyperspectral embeddings. In addition, Li *et al.* [20] first account for domain shift in HSI few-shot classification and propose a novel cross-domain few-shot learning method.

Despite their satisfactory classification results, we consider that intraclass discrepancy and mixed background of HSI data cubes would keep hyperspectral embeddings in the same class

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separate in the learned metric space. These metrics tend to use mixed global representation for similarity comparison of hyperspectral embeddings, which may hinder the hyperspectral pixel from highlighting its crucial local characteristics. More attentions on local characteristics can bring more transferrable information across classes in HSI few-shot classification. To address the above issue, this article proposes a novel meta-learning method for HSI few-shot classification that can exploit the local discriminative information of hyperspectral embeddings and diminish the impact from irrelevant regions. Specifically, we design a deep network using the Earth mover's distance (EMD) as metric for similarity comparison between HSI feature representations, namely, HSEMDNet. Our feature extraction module comprises 3-D convolution and residual blocks. We then opt for EMD to calculate the similarity scores between each query–support pair and assign the query sample to the class with the highest score. When under k-shot classification scenario ($k > 1$), we conduct class prototype learning for each hyperspectral class.

The main contributions of this article are presented as follows.

1) We propose to accomplish the challenging yet less studied HSI few-shot classification with meta-learning strategy. For meta-training, we artificially splice multiple hyperspectral datasets to construct a base dataset for few-shot learning. Meta-learning is performed on the base dataset to help HSEMDNet accumulate transferrable knowledge among different classification tasks. Thus, the proposed method can accurately classify a novel HSI with only a few labeled samples.
2) We propose to formalize HSI classification as an optimal matching problem and develop EMD metric to explore local structure of hyperspectral embeddings. The EMD metric can match local representations to reflect overall distance between two hyperspectral embeddings and diminish the distraction of background.
3) We propose to conduct class prototype learning for each hyperspectral class in the k-shot setting ($k > 1$), which can alleviate the misclassification of query samples caused by intraclass differences on sample-wise level.
4) We select three well-known HSI datasets as novel datasets to verify the classification performance and generalization of the proposed method. The experimental results demonstrate that the proposed method achieves state-of-the-art classification results with only a few labeled samples.

The remainder of this article is organized as follows. Section II presents a review of the related work on HSI classification and few-shot learning. Section III provides the elaboration of our method. Section IV contains extensive experimental results and analyses. Conclusions are finally drawn in Section V.

II. Related Work

A. Traditional Methods and Conventional Deep Methods

Pioneering studies focus on using only spectral information of HSI to conduct classification. Because of redundant spectral information, feature extraction and selection excel in reducing spectral dimension. Representative feature extraction methods are successfully applied to HSI classification, e.g., independent component analysis (ICA) [21] and regularized linear discriminant analysis [22]. Instead of using advanced optimization techniques, Lin *et al.* [23] are in defense of the Iterated Conditional Mode method for HSI classification. Band selection methods [24], [25] are another solution to this problem. With the improvement of HSI resolution, spatial information attracts increasing concern. Recent studies [26], [27] have proven that incorporating spectral and spatial information is conducive to boosting classification performance. In addition, Liu *et al.* [28] propose generalized tensor regression to conduct HSI classification. Whereas these methods extract features in a hand-crafted manner, which heavily lean on prior knowledge. Hence, it is necessary to design an automatic, end-to-end model for hyperspectral feature extraction.

Owing to the end-to-end training manner and superior ability of information processing, deep learning has recently achieved convincing performance for hyperspectral feature representation. Some CNN-based models have boosted the classification accuracy, e.g., diverse region-based CNN [29] and contextual CNN [30]. Hang *et al.* [31] add the popular attention mechanism to CNN. Considering the characteristic of HSI, 3-D convolution is proven more efficient for feature extraction than ordinary 2-D convolution. For this study, Chen *et al.* [32] design a 3-D CNN to simultaneously extract spectral and spatial information. Dwivedi *et al.* [33] propose a novel 3-D CNN with localized residual connections. Moreover, Ge *et al.* [34] develop a 2-D-3-D CNN to merge the features extracted from multiple branches. Liu *et al.* [35] propose a multiscale alternately updated clique network to fully use HSI features in different scales under limited training samples. Other models such as CapsNet [36] have also obtained outstanding performance. With the prevalence of CapsNet, Jiang *et al.* [37] propose Conv-Caps which exploits the advantages of CNN and CapsNet. Recently, attention mechanism has also been widely used in HSI classification. Zhang *et al.* [38] propose a novel spectral–spatial–semantic network with the multiway attention mechanism for HSI classification to extract key informative spectral–spatial features efficiently. Nevertheless, the excellent performance of these models relies excessively on sufficient labeled samples.

B. Semisupervised Learning Methods

As known, acquiring massive labeled HSIs is expensive and inefficient. Semisupervised learning methods have a reputation for alleviating the scarcity of labeled samples, which causes increasing concern in HSI classification. Recently, Hao *et al.* [39] use the clustering method to synthesize pseudo-labels of unlabeled data and feed all data into the designed convolutional recurrent neural network (CRNN) model. Cao *et al.* [40] propose a semisupervised wrapper band selection method that introduces a dynamic classifier to choose the most discriminating bands. Besides, data augmentation is a straightforward solution for the scarcity of hyperspectral samples. For example, Gao *et al.* [41] design a dynamic data augmentation method based on the Siamese structure.

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But these methods tend to learn an efficient classifier for a certain HSI, thus ignoring the correlation between different hyperspectral classes. De facto retraining the model for each novel HSI consumes excessive time and memory, which is infeasible in real applications.

C. Few-Shot Learning Methods

Humans can recognize unseen classes with a few labeled samples based on the knowledge they have learned. Inspired by this, researchers dig deeper into few-shot learning. Similar to transfer learning [42], they aim to enhance the generalization of models in multiple tasks. Optimization approaches represented by [43] aim to learn good initialization and perform fine-tuning on the model with a few labeled samples. Pioneering work, i.e., a model-agnostic meta-learning framework (MAML), uses second-order derivative information to update meta-learning parameters so that model can classify novel classes with several gradient update steps. Recent MAML-based models have constantly refined classification results, e.g., an implicit MAML algorithm (IMAML) [44] and meta-curvature (MC) [45]. Another promising solution is metric-learning approaches, where the general linear classifier head is inherited by a metric function. Matching networks [15] aggregate attention mechanism and cosine distance for similarity measurement. Prototypical network [16] calculates the mean representation for each support class and determines the class membership by Euclidean distance. Relation network [17] contrives a relation module as a learnable metric for obtaining relation scores between each query–support pair. Domain adaptation methods are also emerging. Recently, Lin *et al.* [46] propose a dual adversarial network for domain adaptation, where feature adaptation and classification are conducted iteratively.

These related works from the computer vision area enlighten researchers to introduce few-shot learning to HSI classification. For example, recent studies [18], [19], [47] extend [16], [17] to HSI classification and achieve remarkable performance. General metrics focus on a global representation to compare similarities between hyperspectral embeddings. However, a mixed global representation may withhold the hyperspectral pixel from highlighting its crucial local characteristics. Recently, EMD [48] has been adopted as a metric for few-shot learning [49]. Motivated by this, we propose a novel method for HSI few-shot classification that uses EMD to exploit the local discriminative information of hyperspectral embeddings.

III. Proposed Approach

This section brings an elaboration on our designed method for HSI few-shot classification. We begin with a problem definition of HSI few-shot classification. Then we present the overview and each modules of our proposed method. Finally, we show the detailed algorithm flow.

A. Problem Definition

We first briefly introduce meta-learning mechanism in few-shot learning tasks. The basic setting for few-shot learning is N-way K-shot, where N represents the number of selected classes, and K represents the number of labeled samples in each selected class. Different from general deep methods, meta-learning methods are task-dependent. As can been seen in Fig. 1, the basic unit of the batch has changed from a single image data to a task that consists of a support set and a query set. Following the basic setting of few-shot learning, we denote a task: 1) the support set $S = \{(x_i, y_i)\}_{i=1}^{N\times K}$ and 2) the query set $\mathcal{Q} = \{(x_i, y_i)\}_{i=1}^{N\times Q}$ where x_i and y_i represent a hyperspectral data cube and its label, respectively. Note that these two sets are disjoint with each other.

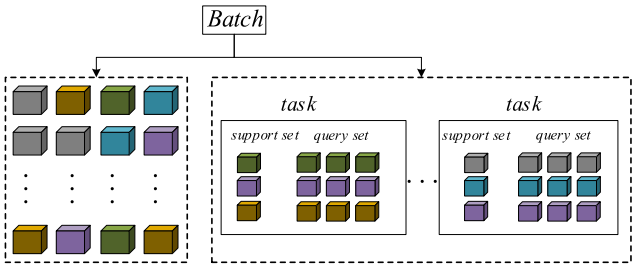


Fig. 1. Batch components of a general deep model (Left) and meta-learning model (Right). Each cube denotes the input sample belonging to each hyperspectral pixel and different colors denote various classes.

The general few-shot datasets such as Mini-ImageNet are divided into two disjoint subsets, i.e., base classes for meta-training and novel classes for meta-testing. The number of base classes is usually much larger than that of novel classes, and thus, meta-learning can better explore the essence of sufficient classification tasks. Currently, there are no few-shot benchmark datasets for HSI classification. Following the widely used strategy [18], [19], [47], we artificially splice multiple hyperspectral datasets to construct the required base dataset and select different HSIs as the novel datasets.

B. Overview of Our Framework

Fig. 2 presents the framework of our proposed meta-learning model for HSI few-shot classification. As illustrated in Fig. 2, we can clearly see how our method works under the meta-learning mechanism. We summarize the elaborate process as follows.

1) *Meta-Training Stage:* Meta-learning is performed on the constructed base dataset to enable the meta-learner to obtain transferrable knowledge from different classification tasks.
2) *Meta-Testing Stage:* When processing a novel HSI, we follow MAML [43] to fine-tune the meta-learner, which ulteriorly updates the model parameters by learning some new tasks from the novel dataset. The purpose of fine-tuning is to help the model quickly adapt to new classification scenarios with individual knowledge from the novel HSI. Finally, our model exploits transferrable knowledge from the base dataset to accomplish novel HSI classification with a few labeled samples.

Specifically, our model uses tasks as input and the label for each query sample as output in an end-to-end manner. We take three-way one-shot classification tasks as an example. Each task consists of any three classes of hyperspectral elements

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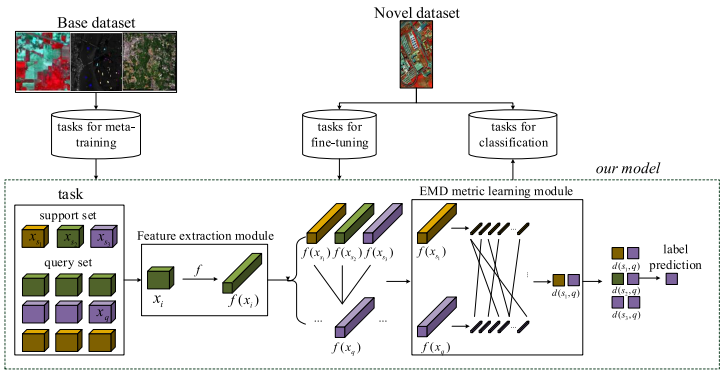


Fig. 2. Overview of our meta-learning method for HSI few-shot classification (three-way one-shot tasks for example). Support and query samples are extracted with designed module f. Then each query–support feature pair is fed through the EMD metric learning module for distance measurement. Label prediction for each query sample is assigned by the minimized distance. Our model is meta-trained and fine-tuned on the base dataset and novel dataset separately, thus accomplishing the final classification on the novel dataset.

in the dataset, where one labeled sample is randomly selected from each selected class to constitute the support set for the task. Feeding a task into the model, our feature extraction module nonlinearly maps all the samples to an embedding space. Then paired embeddings of the query and each support sample are fed into EMD metric learning module for similarity measurement. In essence, we aim to learn a deep embedding space based on EMD that gathers embedding features from the same class and separates embedding features from different classes. The smaller the distance between two embedding features, the more similar they are. Finally, we assign each query sample the class label with the highest similarity score. Sections III-C and III-D detail each designed module in the proposed method. Moreover, we design a class prototype learning module for each hyperspectral class in the support set when coming k-shot classification tasks ($k > 1$), which will be introduced in Section III-E.

C. Feature Extraction Module

Feature extraction module is designed to embed the original hyperspectral data cubes into the feature space, where more discriminative features can be learned. Formally, an initial data cube x_i is nonlinearly mapped to a new feature representation $f(x_i)$. The convolution operation has shown superiority in extracting hyperspectral data features. Considering the redundancy of spectral–spatial information of HSI, this section develops a neural network stacked with the 3-D convolutional layer as the feature extraction module.

To incarnate the advantage of EMD metric to be introduced in the following section, a simple but effective feature extraction module is designed, as shown in Fig. 3. Following [18], the kernel size of the 3-D convolution layers is generally set as $3 \times 3 \times 3$. The batch normalization layer is added after each 3-D convolutional layer to effectively alleviate gradient vanishing and help the model converge faster. A variant of the classic Relu function, i.e., LeakyRelu, is selected as the activation function. We denote the activation function as g, and for the input x, we have

$$g(x) = \max(\alpha x, x) \tag{1}$$

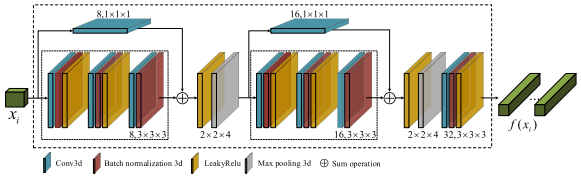


Fig. 3. Architecture of our feature extraction module in HSI few-shot classification. It combines 3-D convolution with basic residual blocks.

where α is a small constant and x is an input vector. In this case, the problem of aliasing in the gradient direction can be avoided due to its nonzero derivative. To alleviate the degradation problem of deep networks, the idea of the residual network [50] is introduced into feature extraction module. Two short-cut connections are added for residual learning. It is noteworthy that before each short-cut connection, a $1 \times 1 \times 1$ 3-D convolution is performed to keep the dimensions of the identity map equal. The $2 \times 2 \times 4$ max-pooling layer is designed for reducing the dimensionality of embedding features such that redundancy of the spectral dimension can dwindle faster.

To sum up, the feature extraction module embeds all the samples from both the support and query sets into the feature space, which serves subsequent metric learning.

D. EMD Metric Learning Module (One-Shot)

EMD [48] investigates to measure the distance between two sets of weighted distributions, which depends on the weight of each unit and the basic distance between units. Given two distributions, one consists of massive Earth in space, and the other involves a collection of holes. Intuitively, the problem solved by EMD is to find the minimized transporting cost to fill in these holes with Earth. The essence of calculating EMD is analogous to find an optimal solution to the transportation problem (TP) in linear programming. To be specific, let $P = \{(p_i, w_{p_i}) | i = 1, 2, \ldots, m\}$ denote a set of suppliers where p_i is the ith supplier and w_{p_i} is the homologous supply units. A set of demanders are denoted as $Q = \{(q_j, w_{q_j}) | j = 1, 2, \ldots, n\}$ where q_j is the jth

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demander and w_{q_j} is the corresponding demand. The cost matrix $C = [c_{ij} | i = 1, 2, \ldots, m, \ j = 1, 2, \ldots, n]$ records each transportation cost c_{ij} from supplier p_i to demander q_j. We aim to ascertain a cheapest flow $\tilde{F} = \{f_{ij} | i = 1, 2, \ldots, m, \ j = 1, 2, \ldots, n\}$ from the suppliers P to the demanders Q:

$$\min_{f_{ij}} \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} f_{ij} \tag{2}$$

subject to $f_{ij} \geq 0$, $\sum_{i=1}^{m} f_{ij} = w_{q_j}$, $\sum_{j=1}^{n} f_{ij} = w_{p_i}$, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$. Here, w_{p_i} and w_{q_j} are the weights of units, which control the total matching flows from each unit. c_{ij} denotes the cost transported from supplier p_i to demander q_j. f_{ij} denotes the number of goods transported. EMD aims to find an optimal matching $\tilde{F}$ between P and Q such that the matching cost is the least. Actually, we can achieve the optimal matching flows by solving the linear programming problem.

Different from conventional metrics such as Euclidean distance or cosine distance, EMD reflects the overall distance between two distributions by matching individual elements. In the existing HSI few-shot classification methods, the similarity of embedding features is commonly reflected with distance between global representations. In contrast, we propose to decompose HSI embedding features into a set of local representations and opt for EMD to obtain the similarity. De facto, exploiting the local representations of features can bring more transferrable information across tasks, which is more in line with the meta-learning mechanism.

Fig. 4 presents our EMD metric learning module, which aims to calculate the distance between each query–support embedding pair. Note that this module is designed for HSI one-shot classification. When the number of support samples per class is greater than 1, we attempt to learn the prototype representation for each hyperspectral class in the support set before metric learning. A detailed state will be given in Section III-E. Given a pair of embedding representations from the support and query set, respectively, we denote as $X \in \mathbb{R}^{C \times D \times H \times W}$ and $Y \in \mathbb{R}^{C \times D \times H \times W}$, where C represents the dimension of embedding features and $D \times H \times W$ denotes the spectral–spatial size of embedding features. We decompose each embedding representation into a set of local feature vectors as $X = [x_1, x_2, \ldots, x_{DHW}]$, where $x_i \in \mathbb{R}^{C \times 1}$ is regarded as a node in X (the same for Y). Thereby two sets of vectors can be used to solve for the lowest matching cost in (2), which reflects the similarity between X and Y. The lower matching cost leads to higher similarity. Cosine similarity is applied to calculate the distance between the nodes in two sets of vectors, denoted as $c(x_i, y_j)$. We denote the matching cost of the two nodes as

$$c_{ij} = 1 - c\left(x_i, y_j\right) \tag{3}$$

which is negatively related to $c(x_i, y_j)$.

As stated in (2), the weight of each node w controls the value of total matching flows from this node. When assigning weights to nodes in X, the dot product of x_i and the mean node feature in Y are defined as w_{x_i}:

$$w_{x_i} = x_i^T \cdot \frac{\sum_{j=1}^{DHW} y_j}{DHW} \tag{4}$$

where DHW is short for $D \times H \times W$ and y_j is any node in Y. To ensure that the weights are nonnegative and the total weights of both the embeddings for matching are equal, we further transform:

$$\hat{w}_{x_i} = \max\left(w_{x_i}, 0\right) \cdot \frac{DHW}{\sum_{j=1}^{DHW} \max\left(w_{x_j}, 0\right)} \tag{5}$$

where w_{x_i} denotes the weight of each node x_i. This cross-reference weighting method focuses on the feature regions that simultaneously appear in paired hyperspectral embeddings and diminishes the influence of absent objects or high-variance background regions on the overall distance. The linchpin to calculate EMD is to solve the linear programming problem of (2) for the optimal matching flows. Fortunately, both the existing GPU-accelerated convex optimization QPTH library and Open computer vision (CV) library with improved simplex algorithm can serve this problem. Accordingly, we can compute the distance d between paired embedding features with

$$d(X, Y) = \sum_{i=1}^{DHM} \sum_{j=1}^{DHM} c_{ij} \tilde{f}_{ij} \tag{6}$$

where $\tilde{f}_{ij}$ denotes the optimal matching flow from node x_i to node y_j. De facto, the similarity score based on (6) can be represented as

$$s(X, Y) = \sum_{i=1}^{DHM} \sum_{j=1}^{DHM} \left(1 - c_{ij}\right) \tilde{f}_{ij} = \sum_{i=1}^{DHM} \sum_{j=1}^{DHM} c\left(x_i, y_j\right) \tilde{f}_{ij}. \tag{7}$$

Finally, the EMD metric learning module produces the similarity scores between the query and each support sample. The support label with the maximum score is assigned to the query sample.

E. Class Prototype Learning Module (k-Shot)

Hyperspectral samples from different classes may behave similar characteristics, which will engender misclassification of query samples on sample-wise level. In the case of few-shot learning ($k > 1$), we conduct HSI classification with prototype computation on class-wise level. De facto, prototype computation can be considered as hard clustering in the support set, where one cluster represents one class and each support sample is assigned to its corresponding class cluster. We attempt to learn a class prototype for each hyperspectral class in the support set which aggregates the sample features from the same class. Classification is then conducted for each query by matching the most similar class prototype representation.

This section presents learning the class prototype with EMD. The prototype is first initialized as the mean embedding feature per class and then updated by the gradient-based optimization algorithm such as Adam [51]. All the support samples are reconstituted into new query sets by a mini-batch

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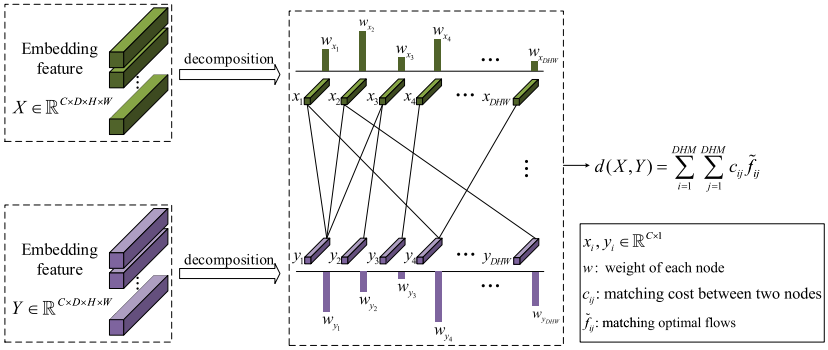


Fig. 4. Visual representation of our EMD metric learning module for paired hyperspectral embedding features. It decomposes each hyperspectral embedding feature into a set of local representation. The EMD metric calculates the distance between two distributions by matching individual elements.

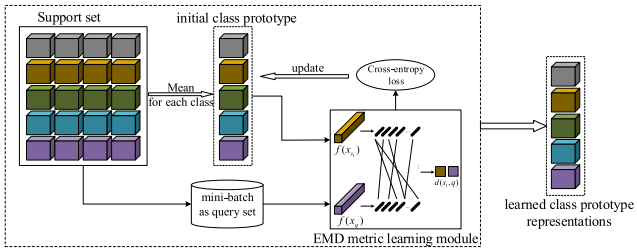


Fig. 5. Visual procedure of our class prototype learning for each hyperspectral class in the support set (five-way four-shot tasks for example). We treat class prototype representations as learnable parameters for training and use the original inputs to construct new tasks. We represent each class by the mean of its samples and use it as initialization. Following the basic setting of tasks, learnable class prototypes are used as support samples of new tasks and new query samples are randomly selected from the original inputs. New tasks are fed into the EMD metric learning module. Cross-entropy loss is applied to update the learnable class prototypes.

form, which are fed into the EMD metric learning module with initial prototype representations. We use the cross-entropy loss to update the class prototype in a batch training manner. After several optimization iterations, we follow Section III-D to calculate the similarity scores between the query and each learned class prototype representation. Fig. 5 shows the detailed learning process.

F. Learning Procedure for Both Meta-Training and Fine-Tuning Stages

The meta-learning mechanism described in Section III-A is used to update our model in meta-training and fine-tuning stages. Cross-entropy is adopted as our loss function:

$$L = -\sum_{i=1}^{N}\sum_{j=1}^{N\times Q} y_j \log\left(s_{ij}\right) \tag{8}$$

where s_{ij} is the similarity scores between the jth query sample and the ith class prototype representation obtained by (7), and y_j is the true label vector of the jth query sample. The complete algorithm process for HSI few-shot classification is illustrated in Algorithm 1.

Algorithm 1 Model Meta-Training or Fine-Tuning for HSI Few-Shot Classification

Require: N-way K-shot task set T sampled from base or novel dataset
(randomly) initialize θ
while not done **do**
  sample batch of tasks t_i from T
  for all t_i **do**
    for all data cube $x \in t_i$ **do**
      input x into feature extraction module in Fig. 3 to obtain embedding feature $f(x)$
    end for
    divide all embedding features into support set S and query set Q
    if $K > 1$ **do**
      conduct class prototype learning on S to get S'
    feed S(or S') and Q into EMD metric learning module to get similarity scores with (7)
    calculate the cross-entropy loss with (8)
  end for
  use the total loss to update θ by Adam optimizer
end while

IV. Experiments and Discussion

A. Experimental Datasets

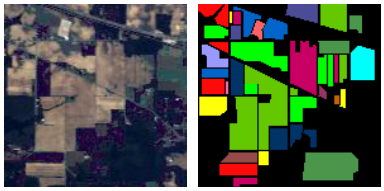
There are currently no specific datasets serving for HSI few-shot classification. Meta-learning on single dataset limits the meta-learner's learning ability. As stated in problem definition section, the HSI few-shot dataset is constructed as follows.

1) Base Dataset: We artificially splice several different HSIs into a new base dataset after standardization. In our experiment, IP, Kennedy Space Center (KSC), and Botswana are selected to construct the base dataset. Table I provides the detailed characteristics of the three HSIs. They provide spectral–spatial information of different ground objects in diverse regions under different ground sample distances. The constructed base dataset contains a total of 43 classes of ground objects, ensuring the diversity of tasks for meta-training. The composition of bands 29, 19, and 9 of three HSIs

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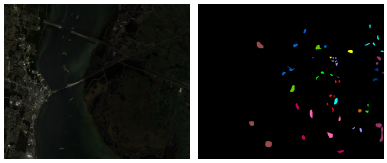
TABLE I
DETAILS OF THREE HSIS FOR BUILDING THE BASE DATASET

	IP	KSC	Botswana
Spatial size (pixel)	145 × 145	512 × 614	1476 × 256
Spectral range (nm)	400-2500	400-2500	400-2500
Number of bands	200	176	145
GSD (m)	20	18	30
Areas	Indiana	Florida	Botswana
Sensor type	AVIRIS	AVIRIS	EO-1
Number of class	16	13	14
Labeled samples	10249	5211	3248



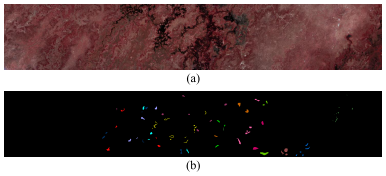
(a) (b)

Fig. 6. IP dataset. (a) Sampled pseudocolor image. (b) Ground-truth map.



(a) (b)

Fig. 7. KSC dataset. (a) Sampled pseudocolor image. (b) Ground-truth map.



(a)

(b)

Fig. 8. Botswana dataset. (a) Sampled pseudocolor image. (b) Ground-truth map.

is extracted to form the sampled pseudocolor images. Formed images and ground-truth maps are shown in Figs. 6–8.

To maintain uniform input size for feature extraction module, we select 100 bands of each pixel via principal component analysis (PCA) [12]. Following [18], [52], the 9×9 neighborhood around each pixel is fed into the network as the spatial information. Therefore, the base dataset composes of pixel data cubes whose size is $9 \times 9 \times 100$.

2) Novel Dataset: To verify the learning capability of the proposed method, three novel HSIs are chosen for meta-testing, including the University of Pavia (UP), the Pavia Center (PC), and the Salinas. The detailed characteristics of novel datasets are presented in Table II. To embody the generalization of the model, we fine-tune the model with five labeled samples of each class in the novel dataset. The remaining samples act as the query sets in the final predictive classification.

TABLE II
DETAILS OF THREE NOVEL DATASETS FOR META-TESTING

	UP	PC	Salinas
Spatial size (pixel)	610 × 340	1096 × 715	512 × 217
Spectral range (nm)	430-860	430-860	400-2500
Number of bands	103	102	204
GSD (m)	1.3	1.3	3.7
Areas	Pavia	Pavia	California
Sensor type	ROSIS	ROSIS	AVIRIS
Number of class	9	9	16
Labeled samples	42776	148152	54129

TABLE III
PARAMETERS OF OUR FEATURE EXTRACTION MODULE. SHORT CONNECTION (SC), MAX POOLING (MP)

Layer	Input Layer	Filter Size	Padding	Output Size
Input	/	/	/	9×9×100
Conv3d-1	Input	1×1×1×8	Y	9×9×100×8
Conv3d-2	Input	3×3×3×8	Y	9×9×100×8
Conv3d-3	Conv3d-2	3×3×3×8	Y	9×9×100×8
Conv3d-4	Conv3d-3	3×3×3×8	Y	9×9×100×8
Sc 1	Conv3d-1 + Conv3d-4			9×9×100×8
Mp-3d 1	Sc 1	2×2×4	N	4×4×25×8
Conv3d-5	Mp-3d 1	1×1×1×16	Y	4×4×25×16
Conv3d-6	Mp-3d 1	3×3×3×16	Y	4×4×25×16
Conv3d-7	Conv3d-6	3×3×3×16	Y	4×4×25×16
Conv3d-8	Conv3d-7	3×3×3×16	Y	4×4×25×16
Sc 2	Conv3d-5 + Conv3d-8			4×4×25×16
Mp-3d 2	Sc 2	2×2×4	N	2×2×6×16
Conv3d-9	Mp-3d 2	3×3×3×32	Y	2×2×6×32
Output	/	/	/	2×2×6×32

TABLE IV
TIME CONSUMPTION OF AN EPISODE (AVERAGE EXECUTION TIME OF TEN RUNS FOR A 20-WAY ONE-SHOT 19-QUERY TASK)

Name	Embedding size	Time
OpenCV	2×2×6×32	3.172s
QPTH	2×2×6×32	8.527s

B. Experimental Setup

Table III lists the detailed parameters of the feature extraction module. The embedding features of size $2 \times 2 \times 6 \times 32$ can be viewed as a set of 32-D vectors and fed into the EMD metric learning module. We compare the execution time of two linear programming solvers in Section III-D, i.e., QPTH and OpenCV. Taking a 20-way one-shot 19-query task as an example, Table IV provides the execution time of the two solvers in an episode. As OpenCV is faster but nondifferentiable, we use QPTH for meta-training and fine-tuning and OpenCV for speeding up classification.

It is important to set optimal N, K, and Q for meta-training. A series of experiments are performed to verify the

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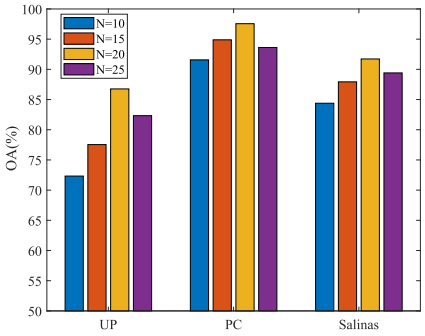


Fig. 9. OA under different N for three novel datasets.

performance with varying N ranging from [10, 15, 20, 25]. Fig. 9 reveals that on all the datasets, overall accuracy (OA) first gradually rises as N increases, and then decreases after N is larger than 20. Thus, this work empirically sets the size of each class, i.e., $N = 20$, in the following experiments. Following [17], 20 samples of each class are randomly selected to construct tasks. We randomly select one labeled sample per class as support samples for training. Formally, we set $K = 1$. The total number of support and query samples for each class in the task is 20, that is, $K + Q = 20$. Our model requires to be fine-tuned when meta-testing, and thus, we only need to meta-train the basic one-shot model ($K = 1$). When $K > 1$, the model has to consume extra time for class prototype learning, which will reduce the training efficiency. Thus, we set $K = 1$, $Q = 19$ for meta-training. For fair comparison, we set the episodes for meta-training as 10 000 and for fine-tuning as 1000 referring to [19]. The learning rate is fixed as 0.001. Moreover, we initialize the weight parameters of our model using the He-initialization method [53] and adopt the Adam optimizer to update our learnable parameters.

C. Comparison Results With Different Methods

The OA, the accuracy of each class, average accuracy (AA), and kappa coefficient (κ) are used to quantitatively analyze the classification performance of the proposed method. Specifically, OA, AA, and κ, respectively, represent the accuracy of all the test samples, the average of each class accuracy, and the classification consistency of all the classes. In our experiment, we compare the proposed method with eight state-of-the-art related methods, including SVM, transductive support vector machine (TSVM) [54], DCGAN [13], HSGAN [14], double-branch multi-attention mechanism network (DBMA) [55], deep few-shot learning method (DFSL) [18], relation network model for HSI few-shot classification (RN-FSC) [19], and deep cross-domain few-shot learning method (DCFSL) [20]. TSVM can be regarded as the semisupervised extension from SVM. DCGAN and HSGAN can exploit implicit information from unlabeled samples. DBMA designs two attention mechanisms for spectral and spatial information of HSI, and thus outperforms most CNN-based methods. DFSL, RN-FSC, and DCFSL are the meta-learning methods for HSI few-shot classification. DFSL uses an SVM classifier to classify the embedding features. RN-FSC develops a relation learning module to produce relation scores between samples. DCFSL adopts a conditional adversarial domain adaptation strategy to achieve domain distribution alignment.

Note that for all the compared methods, only five labeled samples per class on novel datasets are randomly selected for supervision learning. Nonmeta-learning methods, i.e., SVM, TSVM, DCGAN, HSGAN, and DBMA, are directly trained on novel datasets. Five labeled samples per class are used for training and the rest samples are used for testing. Meta-learning methods, i.e., DFSL, RN-FSC, DCFSL, and HSEMD-Net, are meta-trained on the base datasets and fine-tuned on novel datasets. Five labeled samples per class are used for fine-tuning. Tables V–VII present the quantitative classification results of all the methods on three novel datasets, from which five conclusions can be summarized as follows.

1) Among all the comparison methods, SVM and TSVM perform worst on three novel datasets. When the number of labeled samples per class is few, the advantage of TSVM is inadequate. For example, the OA of TSVM on the UP dataset exceeds that of SVM by 7.13% but underperforms SVM on the PC and Salinas datasets.
2) HSGAN and DCGAN achieve similar performance and outperform SVM on three novel datasets. HSGAN and DCGAN improve OA by 9.85% and 8.74% over SVM on the PC datasets. The superiority of HSGAN and DCGAN is to incorporate unlabeled samples for end-to-end training.
3) DBMA further improves the OA by 5.6% over DCGAN and by 5.41% over HSGAN on the UP dataset. But the OA of DBMA is slightly inferior to that of HSGAN on the PC dataset. Under the few-shot setting, DBMA achieves comparable performance with the semisupervised methods.
4) The OA of DFSL, RN-FSC, and DCFSL further improves on three novel datasets. DCFSL always performs better than DFSL and RN-FSC. DCFSL improves 4.53% and 4.56% over DBMA on the UP and PC datasets, respectively. However, DFSL underperforms DBMA on the UP and Salinas datasets.
5) The proposed HSEMDNet shows the most competitive performance with the highest OA, AA, and κ on three novel datasets. For instance, it, respectively, improves the OA by 3.93% and 2.44% over DCFSL on the UP and Salines datasets. The AA of the proposed HSEMDNet rises by 4.17%, 4.23%, and 1.08% than that of RN-FSC on three novel datasets. However, AA of DCFSL rises by 0.96% than that of the proposed HSEMDNet. In addition, κ of HSEMDNet is 7.5% higher than that of RN-FSC on the UP dataset.

In addition, Figs. 10–12 present the classification maps on three novel datasets to intuitively compare the classification results of all the methods. Consistent with the qualitative results, we conclude that with the rise of OA, the difference between the classification map and ground truth shows a decreasing trend. Visually, the classification maps of the proposed model on three novel datasets behave best among all the methods, and thus, combining the quantitative results in Tables V–VII confirms the effectiveness of our method.

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TABLE V

CLASSIFICATION RESULTS OF COMPARISON METHODS ON UP (FIVE LABELED SAMPLES PER CLASS USED AS THE SUPERVISED SAMPLES; BOLD VALUES REPRESENT THE BEST RESULTS)

Class	SVM	TSVM	DBMA	DCGAN	HSGAN	DFSL	RN-FSC	DCFSL	HSEMDNet
Asphalt	95.56	96.55	96.97	92.18	96.94	78.05	87.28	82.20	93.34
Meadows	80.54	80.47	93.16	90.32	92.19	88.54	84.33	87.74	80.17
Gravel	29.36	11.11	57.43	41.80	52.30	57.40	90.42	67.46	86.52
Trees	59.95	48.71	89.82	86.39	99.22	62.73	78.09	93.16	71.77
Metal Sheets	93.58	94.92	99.92	83.30	100.00	98.82	99.56	99.49	100.00
Bare Soil	26.94	37.91	47.62	43.63	38.34	43.26	63.25	77.32	66.67
Bitumen	31.23	20.50	80.36	44.54	57.20	100.00	52.09	81.18	82.70
Bricks	54.27	55.36	63.96	62.11	90.14	71.66	84.81	66.73	92.34
Shadows	98.91	99.89	92.98	66.33	48.10	97.83	95.94	98.66	99.75
OA (%)	54.79	61.92	79.12	73.52	73.71	73.51	81.94	83.65	**87.58**
AA (%)	63.37	60.60	80.24	67.84	74.94	77.57	81.75	83.77	**85.92**
κ (%)	49.64	51.44	71.59	66.07	67.16	66.21	75.84	78.70	**83.34**

TABLE VI

CLASSIFICATION RESULTS OF COMPARISON METHODS ON PC (FIVE LABELED SAMPLES PER CLASS USED AS THE SUPERVISED SAMPLES; BOLD VALUES REPRESENT THE BEST RESULTS)

Class	SVM	TSVM	DBMA	DCGAN	HSGAN	DFSL	RN-FSC	DCFSL	HSEMDNet
Water	99.95	95.12	98.99	98.13	97.99	88.46	100.00	99.77	100.00
Trees	94.68	92.22	98.76	98.15	98.82	98.08	99.53	90.55	99.47
Meadows	40.86	40.12	78.91	65.81	67.78	80.97	67.60	94.06	84.29
Bricks	56.47	8.12	65.64	55.64	64.42	70.28	72.43	91.42	72.31
Bare Soil	19.51	27.15	64.53	53.42	59.96	78.61	96.91	91.13	98.42
Asphalt	63.66	46.87	86.87	84.21	84.31	85.09	85.86	96.70	90.56
Bitumen	78.21	1.38	94.18	99.37	92.78	100.00	85.55	85.01	100.00
Tiles	88.66	97.14	100.00	99.02	99.93	99.72	99.94	98.89	99.87
Shadows	99.76	93.17	87.62	77.46	80.62	90.36	91.87	98.90	92.93
OA (%)	83.11	67.60	92.89	91.85	92.96	94.07	96.36	97.45	**97.56**
AA (%)	71.31	55.70	86.17	81.24	82.95	87.95	88.86	**94.05**	93.09
κ (%)	76.62	56.60	92.53	91.02	92.18	93.84	95.98	96.40	**96.81**

TABLE VII

CLASSIFICATION RESULTS OF COMPARISON METHODS ON SALINAS (FIVE LABELED SAMPLES PER CLASS USED AS THE SUPERVISED SAMPLES; BOLD VALUES REPRESENT THE BEST RESULTS)

Class	SVM	TSVM	DBMA	DCGAN	HSGAN	DFSL	RN-FSC	DCFSL	HSEMDNet
Brocoli_green_weeds_1	85.60	80.50	86.39	56.94	59.35	87.33	99.26	99.40	97.68
Brocoli_green_weeds_2	98.54	98.08	99.87	71.53	98.88	99.68	100.00	99.76	99.59
Fallow	65.38	65.19	83.36	87.44	77.24	79.97	97.87	91.96	99.86
Fallow_rough_plow	95.82	95.46	97.72	76.45	93.30	97.00	99.50	99.55	100.00
Fallow_smooth	95.83	64.25	97.18	94.95	86.95	96.77	97.81	92.70	98.99
Stubble	99.92	99.95	99.85	99.47	99.90	99.92	99.35	99.52	100.00
Celery	95.29	85.10	97.67	89.63	84.94	94.87	100.00	98.88	100.00
Grapes_untrained	57.00	44.29	68.96	70.93	68.06	70.52	66.24	74.57	64.32
Soil_vinyard_develop	90.64	74.06	98.83	92.89	91.43	97.95	97.34	99.59	98.21
Corn_senesced_weeds	85.87	64.71	92.38	63.58	93.86	83.97	93.66	86.42	94.72
Lettuce_romaine_4wk	38.32	47.56	70.54	83.81	49.04	69.36	73.96	96.61	86.80
Lettuce_romaine_5wk	87.56	92.56	97.66	97.33	88.50	96.21	99.84	99.93	94.87
Lettuce_romaine_6wk	88.66	47.87	90.86	97.53	45.48	96.41	100.00	99.30	98.34
Lettuce_romaine_7wk	87.87	86.81	94.57	87.09	90.34	95.69	96.39	98.85	98.37
Vinyard_untrained	33.18	32.31	62.79	74.78	54.62	66.95	68.85	75.38	77.81
Vinyard_vertical	81.64	54.24	94.77	77.17	60.71	95.26	99.89	92.22	97.64
OA (%)	73.64	64.63	86.33	80.11	75.66	85.36	86.99	89.34	**91.78**
AA (%)	80.45	70.81	89.59	82.60	77.66	89.24	93.12	94.04	**94.20**
κ (%)	70.70	60.95	83.73	77.86	72.90	83.68	85.44	88.17	**89.95**

D. Investigation of the Number of Labeled Samples

This experiment explores the influence of the number of labeled samples on classification performance. Specifically, 5, 10, 15, 20, and 25 labeled samples per class are randomly selected as support samples for DFSL, RN-FSC, DCFSL, and HSEMDNet. They act for fine-tuning. For other nonmeta-learning methods, they act for supervised training. The OA of different methods on three novel datasets is illustrated in Fig. 13.

We observe that the OA of all the methods constantly rises as the number of labeled samples per class increases. Even if

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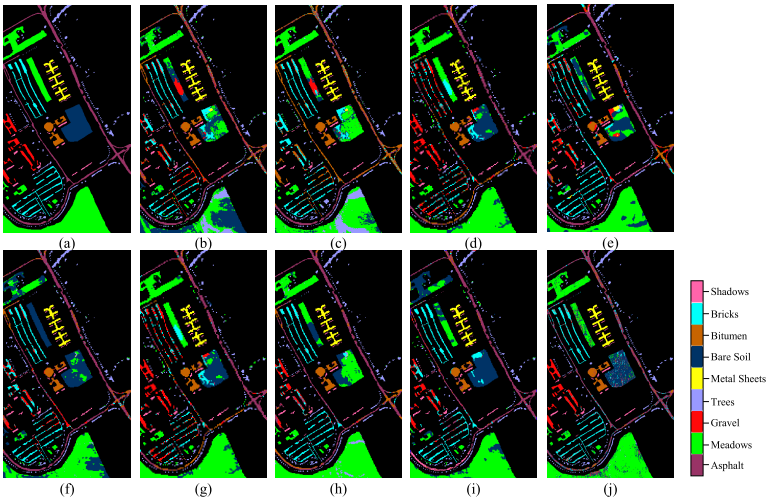


Fig. 10. Classification maps of different methods on the UP dataset. (a) Ground truth. (b) SVM. (c) TSVM. (d) DBMA. (e) DCGAN. (f) HSGAN. (g) DFSL. (h) RN-FSC. (i) DCFSL. (j) HSEMDNet.

we use 25 labeled samples per class for training, the number of supervised samples is still tiny. Thus, the OA curves of SVM and TSVM are always on the bottom. With the number of supervised samples increasing, the OA curve of DBMA gradually converges or even overlaps with that of DFSL and RN-FSL. The OA curve of DCFSL keeps over that of DFSL and RN-FSL on three novel datasets. DCFSL and the proposed HSEMDNet almost keep overlapping curves on the PC dataset. The OA curve of HSEMDNet is always highest on the UP and Salinas datasets, which indicates our method consistently performs best. This advantage is more evident with only five labeled samples per class.

E. Ablation Study

This section explores the influence of three modules on classification accuracy. Note that we fix other settings when analyzing the impact of one module.

1) Effectiveness of Feature Extraction Module: To verify the effectiveness of 3-D convolution and residual learning in the feature extraction module, we compare it with basic fully 2-D CNN and 3-D CNN for hyperspectral feature extraction. Other parameter settings of the two comparison networks follow Table III. We take the UP dataset for instance and record the OA of three feature extraction modules under different K in Table VIII. A gradual rise in OA from 2-D CNN to ours demonstrates that 3-D convolution is more plausible for hyperspectral feature extraction. The OA of ours is roughly 5%–6% higher than that of fully 3-D CNN, which confirms the effectiveness of two basic residual blocks. Under the condition without designing a more complex network structure, our feature extraction module has enabled HSEMDNet to perform satisfyingly.

2) Effectiveness of EMD Metric Learning Module: We verify the effectiveness of EMD metric by replacing it with the Euclidean distance, cosine distance, and dot product which focus on global information. For a fair comparison, we use the same feature extraction module to acquire the hyperspectral embeddings. It is noteworthy that when using the alternatives, we add a global average pooling operation after the feature extraction module. Class prototype learning is unnecessary in one-shot scenarios, and thus, a fair comparison on different metrics can be performed. Table IX details the comparison of OA under four metrics for one-shot classification on three novel datasets. Consistent with theory analysis, HSEMDNet outperforms the other three.

3) Effectiveness of Class Prototype Learning Module: When the number of support samples per class is greater than 1 ($K > 1$), we propose to learn the prototype representations for each hyperspectral class in the support set. We select two alternative methods of class prototype learning to compare classification performance. One alternative is that we directly use the nearest neighbor (NN) classifier to classify query samples on sample-wise level without class prototype learning. Following [16], we average the hyperspectral embeddings of

TABLE VIII

OA (%) UNDER DIFFERENT K ON THE UP DATASET. K REPRESENTS LABELED SAMPLES PER CLASS. BOLD VALUES REPRESENT THE BEST RESULTS

	K = 5	K = 10	K = 15
fully 2-D CNN	77.58	79.64	82.51
fully 3-D CNN	81.23	82.79	85.46
HSEMDNet	**87.58**	**89.33**	**91.72**

TABLE IX

OA (%) UNDER USING DIFFERENT METRICS FOR ONE-SHOT CLASSIFICATION ON THREE NOVEL DATASETS. BOLD VALUES REPRESENT THE BEST RESULTS

Metric	UP	PC	Salinas
Euclidean distance	63.71	77.98	69.42
Cosine distance	65.91	77.64	71.53
Dot product	62.94	78.23	69.19
HSEMDNet	**69.98**	**80.67**	**73.26**

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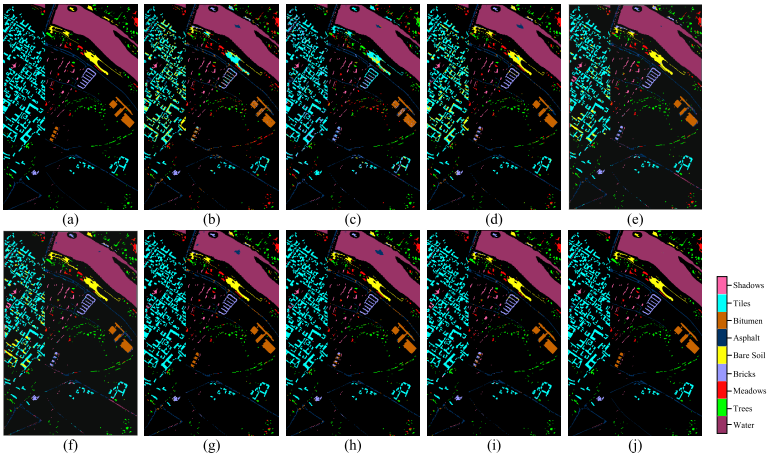


Fig. 11. Classification maps of different methods on the PC dataset. (a) Ground truth. (b) SVM. (c) TSVM. (d) DBMA. (e) DCGAN. (f) HSGAN. (g) DFSL. (h) RN-FSC. (i) DCFSL. (j) HSEMDNet.

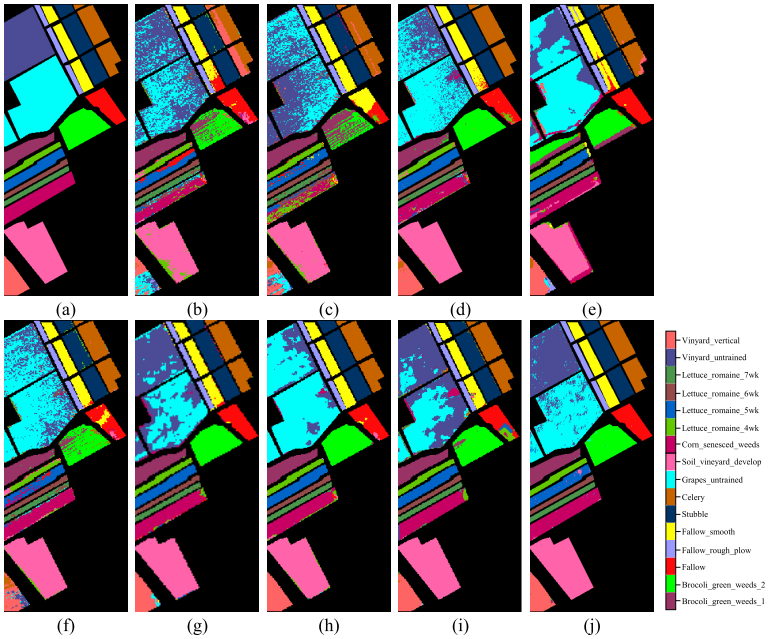


Fig. 12. Classification maps of different methods on the Salinas dataset. (a) Ground truth. (b) SVM. (c) TSVM. (d) DBMA. (e) DCGAN. (f) HSGAN. (g) DFSL. (h) RN-FSC. (i) DCFSL. (j) HSEMDNet.

support samples in each class as the prototype and apply the NN method for classification. We denote this alternative as Avg + NN. Our method is denoted as Ours + NN. Fig. 14 presents the curve of OA on three novel HSIs. It shows a common law that the curves of three methods are on the rise as the number of labeled samples increases. The performance of our prototype learning method is significantly higher than the other two by 3%–5%. Directly using the NN classifier underperforms Avg + NN, which implies that different classes may have the same characteristics in HSI. Our superiority over Avg + NN is that we attempt to learn a more discriminative class prototype based on the mean embedding features. Nevertheless, the curve of NN gradually converges to that of Avg + NN when K increases from 5 to 25 on the PC dataset. To sum up, a good class prototype representation is informative for HSI classification.

F. Investigation of Meta-Training and Fine-Tuning

To investigate the influence of meta-training and fine-tuning on classification accuracy, we design the following combined experiments consisting of three cases: with only meta-training, with only fine-tuning, and with both of them. The results are recorded in Table X, where the checkmark illuminates that the related operation is chosen.

We conclude that meta-training and fine-tuning can greatly improve the final classification accuracy. Specifically, when $K = 5$, meta-training increases the OA by 15.25%, 10.67%, and 14.50% on the UP, PC, and Salinas datasets, respectively. It is obvious that using only a few labeled samples from the novel dataset without meta-training leads to poor learning ability. Fine-tuning further boosts the OA by 18.99%, 17.09%, and 20.02% on the UP, PC, and Salinas datasets, respectively.

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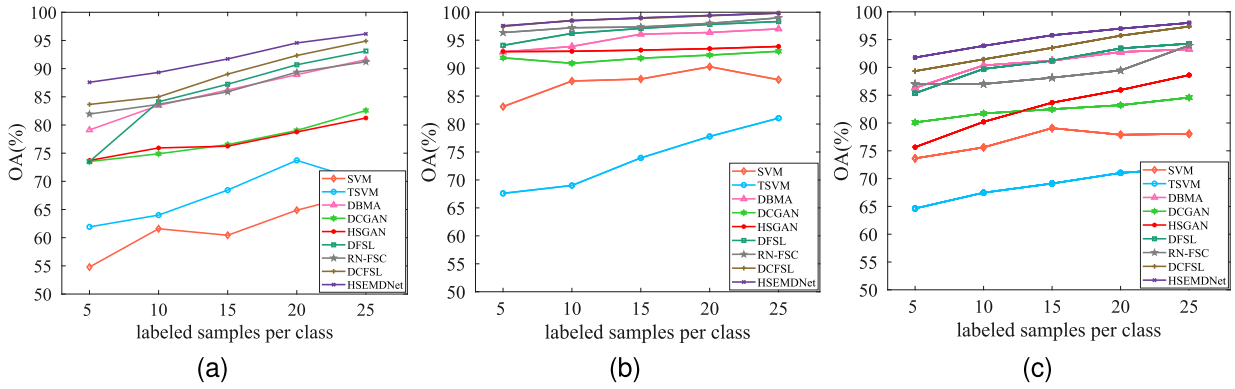


Fig. 13. OA (%) of different methods on three novel datasets with different numbers of labeled samples per class. (a) UP. (b) PC. (c) Salinas.

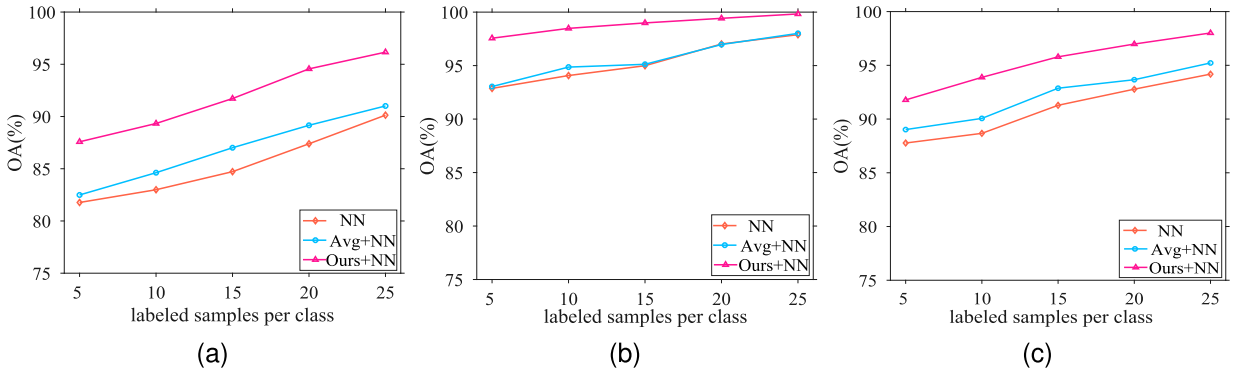


Fig. 14. OA (%) under different K for three methods on three novel datasets. (a) UP. (b) PC. (c) Salinas.

TABLE X

INFLUENCE OF THE META-TRAINING AND FINE-TUNING ON CLASSIFICATION ACCURACY OA (%). K REPRESENTS LABELED SAMPLES PER CLASS. BOLD VALUES REPRESENT THE BEST RESULTS

Novel dataset	Meta-training	Fine-tuning	K=5	K=10	K=15	K=20	K=25
UP	×	√	72.33	76.18	81.09	86.64	90.73
	√	×	68.59	70.98	73.61	76.03	80.84
	√	√	**87.58**	**89.33**	**91.72**	**94.56**	**96.16**
PC	×	√	86.89	88.05	89.87	91.93	94.18
	√	×	80.47	82.72	83.86	85.24	86.31
	√	√	**97.56**	**98.48**	**98.99**	**99.42**	**99.83**
Salinas	×	√	77.28	79.55	81.97	84.42	88.18
	√	×	71.76	73.07	75.26	76.89	78.54
	√	√	**91.78**	**93.88**	**95.79**	**96.98**	**98.01**

It is significant as the model without fine-tuning can hardly learn individual knowledge from the novel dataset. Our model achieves the highest classification accuracy due to the use of meta-training and fine-tuning.

Moreover, our advantages slightly decline as the number of labeled samples per class increases. When the number of labeled samples per class increases from 5 to 25, compared with the model only using fine-tuning, the OA advantage of our model on the UP dataset decreases from 15.25% to 5.43%. Likewise, the OA drops from 10.67% to 5.65% on the PC dataset and from 14.5% to 9.83% on the Salinas dataset. It verifies the importance of individual information on classification accuracy and confirms the validity of meta-training and fine-tuning for our model, especially when the number of labeled samples per class is few.

G. Discussion

For the existing deep models for HSI classification, their performance relies on the number of labeled samples. When performing HSI classification under the few-shot setting, they are prone to overfitting. Hence, we introduce the meta-learning mechanism to the proposed hyperspectral model under the few-shot setting. We perform the comparison and analysis experiments on related hyperparameters and select the optimal parameter combination settings. The proposed model achieves the best classification results on three novel datasets. As the number of labeled samples per class increases, our model still possesses certain competitiveness.

The effectiveness of the three designed modules is investigated through ablation experiments. First, the experimental

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results illustrate that our designed feature extraction module composed of 3-D convolution and two basic residual blocks is simple but effective. Second, the EMD metric pays more attention to the correlation between the local representations of paired hyperspectral embeddings, thus underpinning the more plausible similarity comparison. Hyperspectral samples from the same class may also present different characteristics, which aggravates the misjudgment of classification on sample-wise level. Third, we learn a class prototype for each hyperspectral class to aggregate the embedding features from the same class in the metric space.

In addition, the meta-learning strategy supplies model with great generalization, which helps the model quickly adapt to classification tasks from the novel datasets under the few-shot setting. The model accumulates general knowledge from the base dataset and transfers learned knowledge to classification tasks on the novel dataset. The experimental results demonstrate that meta-training and fine-tuning are of great significance on classification accuracy.

V. Conclusion

This article proposes a novel meta-learning method for HSI few-shot classification that achieves outstanding performance with only a few labeled samples. The EMD into similarity comparison between paired hyperspectral embeddings allows the model to pay more attention to local information and enhance generalization. We further conduct class prototype learning for each hyperspectral class in the support set. The proposed method works under meta-learning mechanism, which learns general knowledge from the base dataset and uses transferrable knowledge to classify a novel HSI. Extensive experiments and analyses on three novel HSI datasets demonstrate that our method achieves highly competitive performance and the meta-learning mechanism shows feasible latent in HSI classification under the few-shot setting.

In the proposed method, learning prototype for each hyperspectral class is time-consuming for inference. Our future work will explore optimizing class prototype without gradient-based updating to improve efficiency.

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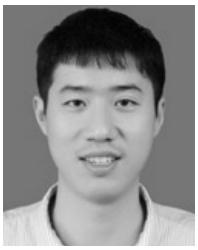
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